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Automatic Detection of Craters and Boulders from Orbiter High-Resolution Camera (OHRC) Images Using YOLO Models

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***Abstract*— This work investigates the automated detection of lunar craters and boulders using advanced Computer Vision (CV) techniques, specifically leveraging the YOLOv11 and YOLO-NAS object detection models on high-resolution images from ISRO's Chandrayaan-2 Orbiter High-Resolution Camera (OHRC). Recognizing the importance of lunar crater studies for understanding impact history, geological evolution, and planning future missions, this research aims to develop a more efficient and scalable solution for identifying and analyzing lunar surface features. The primary objectives include training the YOLOv11 and YOLO-NAS models on OHRC data to detect craters and boulders, evaluating model accuracy using metrics such as mean Average Precision (mAP), and refining detection parameters to optimize performance.**

***Index Terms*—Craters, boulders, object detection, YOLOv11, YOLO-NAS, Chandrayaan-2, planetary exploration, computer vision, AI, machine learning.**

I. Introduction

Scientists have long studied the Moon as a source of information in the study of the history and evolution of the Solar System. The Moon is, after all, Earth's closest neighbor, and a natural laboratory, where most surface data, in the form of craters, contains critical historical information about impacts of meteors, volcanic activities, and surface transformations which reveal forces that have operated on the Moon and on other planetary bodies for billions of years.

Recent developments in space exploration, such as the Chandrayaan-2 mission of India, have facilitated the study of the moon using high-resolution images [1], [2]. However, the massive and complex datasets thus generated are a big challenge to be processed manually. The current project mitigates the above drawbacks by using highly advanced machine learning models-YOLOv11 and YOLO-NAS-for the detection and classification of craters and boulders. These models will be trained on such curated datasets of OHRCs, and their performance optimized for improving the efficiency and accuracy of lunar surface analysis to pave the way for further scientific exploration.

II. Literature Review

Lunar crater and boulder detection becoming increasingly important for the planning of missions and the detection of hazards on the lunar surface, it has become an increasingly important part of space research. The origins of developing automated methods to detect craters can be traced back to early studies into remote sensing and high-resolution satellite images. With deep learning, YOLO (You Only Look Once) has emerged as an incredibly powerful tool in terms of detection, especially for lunar craters, significantly improving detection speed and accuracy [4], [5]. From improvement in YOLO to multi-source data integration, deformable convolutional networks, and attention mechanism integration into feature extraction stages to improve detection capabilities, advancements in the technology have made YOLO-based models more competent to handle complex lunar terrain data.

The newer YOLO models have shown enhanced development of crater detection models. Techniques that have improved the accuracy in the detection process include integration with Digital Elevation Models (DEMs) into deep learning and hybrid techniques involving the combination of YOLO with terrain analysis, ensuring reliability across diversified lunar surfaces. Newer versions of YOLO have been developed, and new architecture searches have been developed through the introduction of YOLO-NAS making the models more efficient and adaptive. All of these advancements have made the detection of lunar craters possible and have even facilitated their application in autonomous systems, such as in navigating a lunar rover. There is an increasing convergence between deep learning and technologies regarding space exploration.

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III. METHODOLOGY

This project follows a structured approach as shown in Figure 1. Each phase, from data preprocessing to model training, is essential for achieving accurate crater and boulder detection.

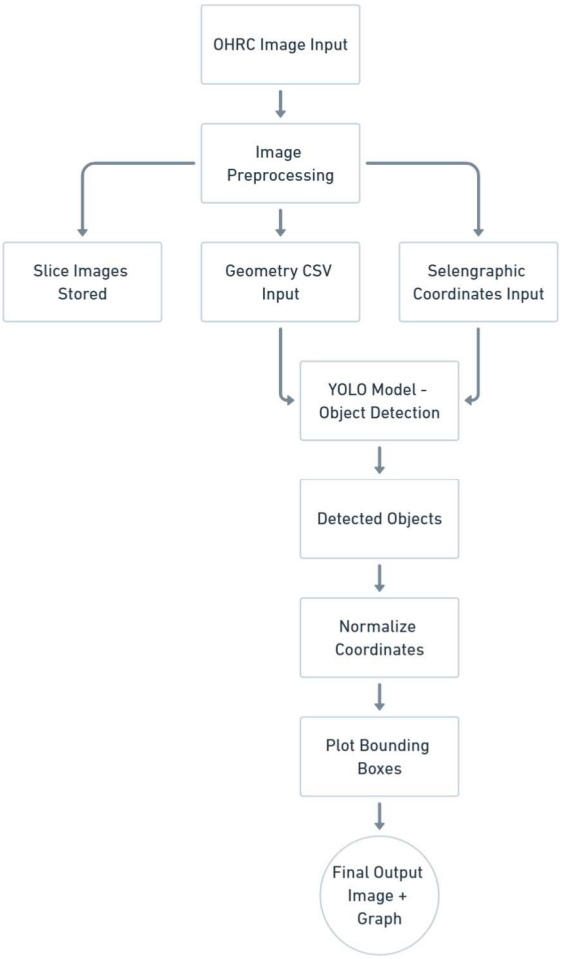


Fig. 1. Methodology Flowchart

A. Dataset Description with Technical Details

- **Source:** The dataset is derived from the OHRC (Orbiter High-Resolution Camera) onboard ISRO's Chandrayaan-2 mission. OHRC captures detailed lunar surface images, offering unparalleled spatial resolution for geological analysis [1].
- **Dataset Size:** Comprises 208 high-resolution grayscale images. The data volume requires efficient storage and processing techniques due to the high pixel density and metadata integration [2].
- **Resolution:** Each image has a resolution of 163x1280 pixels. The horizontal orientation provides an expansive field of view, suitable for detailed surface feature detection. The high aspect ratio demands slicing into smaller, manageable patches for computational efficiency [2].
- **Applications:**
  - **Primary Focus:** Detecting craters and boulders for geological studies [1].
  - **Broader Applications:** Surface mapping for mission planning, hazard zone identification for rover navigation, and resource exploration (e.g., locating water ice in shadowed craters).
- **Structure:**
  - **Data Organization:** Includes labeled bounding boxes for supervised learning (if applicable). Metadata contains selenographic coordinates (latitude and longitude) for spatial mapping of detected features [1].
  - **Division of Dataset:** Training Set (70%), Validation Set (15%), and Test Set (15%) for model development and evaluation [1].
  - **Labeling:** Annotations in standard formats (e.g., YOLO or COCO format) include bounding box coordinates, class labels (crater, boulder), and confidence scores.
- **Preprocessing Requirements:**
  - **Image Slicing:** Divide large images into patches (e.g., 256x256 pixels) to match GPU memory constraints [2].
  - **Normalization:** Scale bounding box coordinates to [0, 1] for compatibility with YOLO models.
  - **Augmentation:** Introduce variations such as rotation, flipping, and brightness adjustments to improve model generalization [1].
  - **Data Cleaning:** Address issues like missing metadata, coordinate misalignments, or inconsistent resolutions.

B. Preprocessing

Comprehensive preprocessing is critical to ensuring the OHRC dataset and related project components are optimized for training YOLOv11 and YOLO-NAS models. This phase integrates multiple steps to prepare the data, align metadata, and establish a robust pipeline for object detection tasks. Key steps include:

- **Dataset Acquisition and Inspection:** - High-resolution OHRC images (163x1280 pixels) were sourced from ISRO's Chandrayaan-2 mission. - Initial inspection was performed to ensure dataset completeness, consistency, and quality. - Any corrupted or incomplete files were logged and excluded from the pipeline.
- **Slicing into Manageable Patches:** - Due to the high resolution of OHRC images, they were divided into smaller, computationally manageable patches

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(e.g., 256x256 pixels) using custom slicing algorithms [11]. - Each patch retained its spatial metadata, ensuring that selenographic coordinates (latitude and longitude) were accurately mapped to the lunar surface.

- **Bounding Box Normalization:** - Annotated bounding boxes for craters and boulders were scaled to a normalized [0, 1] range relative to the dimensions of the image patches. - This standardization ensures compatibility with YOLO models, which require normalized coordinates for object detection.
- **Data Augmentation:** - To improve model generalization, various augmentation techniques were applied, including:
  - Rotation and flipping to simulate different viewing angles.
  - Brightness and contrast adjustments to account for variations in lunar surface lighting.
  - Random cropping and scaling to enhance robustness against size and positional variations [12].
- **Verifying Spatial Integrity:** - Metadata files containing selenographic coordinates were cross-verified with the corresponding image patches to maintain alignment. - A coordinate mapping function was implemented to accurately localize detected craters and boulders in their original positions on the lunar surface.
- **Model Input Preparation:** - Processed patches and normalized bounding boxes were converted into YOLO-compatible formats, including annotation files (e.g., YOLO TXT or COCO JSON). - Training, validation, and test datasets were split (70%, 15%, 15%) to ensure a balanced and unbiased evaluation pipeline.
- **Pipeline Integration:** The preprocessing outputs were directly integrated into the training pipeline, ensuring seamless data loading and augmentation during training. Custom Python scripts and libraries (e.g., OpenCV, NumPy, and PyTorch) were employed for automation and optimization.

C. Preprocessing for Metadata and Results Interpretation

Beyond dataset preparation, preprocessing was also applied to interpret model outputs and ensure scientific accuracy:

- **Coordinate Denormalization:** - Detected bounding boxes from the model outputs were denormalized to real-world pixel dimensions [11]. - Mapped coordinates were then reassembled with their original selenographic positions for scientific analysis.
- **Result Visualization:** - Bounding boxes and detection results were overlayed on the original OHRC images using visualization tools [12]. - Custom scripts were developed to highlight detected craters and boulders, class labels and confidence scores.

D. Model Training

The training process for YOLOv11 and YOLO-NAS models is designed to optimize performance for detecting lunar craters and boulders. Key steps include:

- **YOLO-NAS Architecture (Part 1):** The QSP and QCI blocks are advanced feature extraction modules within the YOLO-NAS architecture. The QSP Block utilizes multiple stacked ZXQARepVGG modules, followed by concatenation and a CBR (Conv-BatchNorm-ReLU) 1×1 layer to enhance feature learning. The QCI Block introduces an additional CBR 1×1 shortcut connection, improving gradient flow and enabling better feature refinement. These blocks contribute to improved model efficiency by optimizing feature representation and reducing redundancy in learned parameters. The YOLO-NAS architecture, with its advanced QSP and QCI blocks, has demonstrated superior performance in feature extraction and object detection tasks [4], [5].

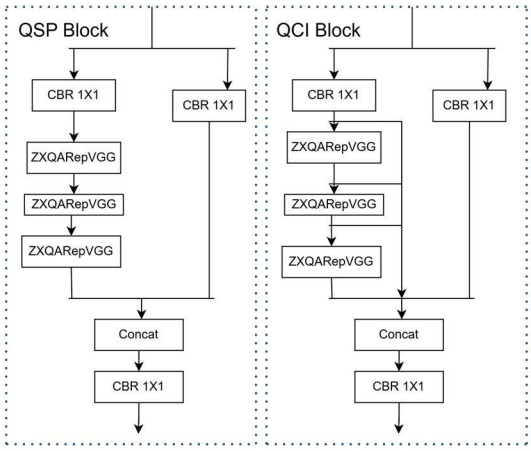


Fig. 2. QSP and QCI Blocks in YOLO-NAS Architecture

- **YOLO-NAS Architecture (Part 2):** The YOLO-NAS architecture comprises three primary components: the Backbone, Neck, and Head. The Backbone extracts hierarchical feature representations from the input image through multiple stages, incorporating convolutional layers and downsampling operations. The Neck facilitates multi-scale feature fusion using a combination of upsampling (Up Stage 5 & 6) and downsampling (Down Stage 7 & 8) pathways, enhancing spatial and contextual information. The Head consists of three detection layers responsible for predicting object classes, bounding

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boxes, and confidence scores at different feature scales. A width multiplier of 0.75 is applied to optimize computational efficiency while maintaining detection accuracy.

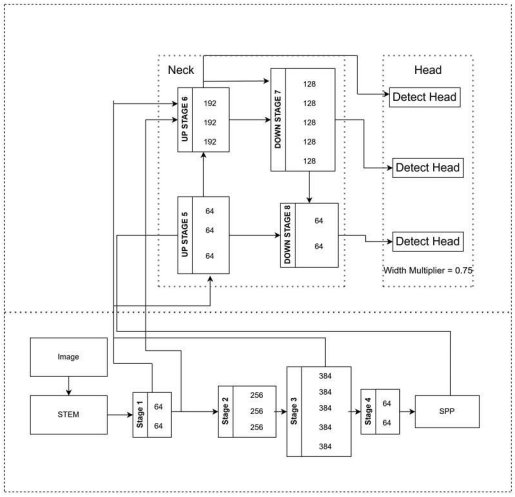


Fig. 3. Backbone, Neck, and Head in YOLO-NAS Architecture

1) Training Parameters: The following parameters were carefully tuned to ensure effective model training:

- **Batch Size**: Determines the number of images processed per iteration. It was adjusted to effectively balance GPU memory consumption and training speed [3].
- **Learning Rate**: Set to an initial value of 0.001, ensuring gradual and stable weight updates for reliable convergence without oscillations [3].
- **Epochs**: Fixed at 300 to allow sufficient iterations over the data set for optimal model generalization.

2) Loss Functions: A composite loss function was applied, comprising the following components:

- **Localization Loss**: Measures the accuracy of bounding box predictions relative to the ground-truth positions.
- **Confidence Loss**: Assesses the confidence of the model in detecting objects within an image.
- **Classification Loss**: Ensures correct identification of object types (e.g., craters and boulders).

3) Fine-Tuning: The models were fine-tuned to address the unique characteristics of lunar data, such as:

- Variations in **crater scales** and **irregular shapes**.
- High contrast features inherent in **grayscale imagery**.
- **Smaller boulders** that pose additional detection challenges.

4) Validation: Model validation was critical to evaluating the performance of YOLOv11 and YOLO-NAS in the detection of lunar craters and boulders. The key aspects are outlined below:

5) Validation Dataset: A separate subset that constitutes 15% of the total dataset was reserved for validation. This step ensures an unbiased performance evaluation, independent of the training process.

6) Evaluation Process: The model predictions (bounding boxes) were compared with **ground truth labels**. Predictions were evaluated based on **IoU thresholds** (e.g., 0.5) to determine correctness.

7) Error Analysis: An in-depth analysis of false positives and missed detections was performed. Specific challenges identified included difficulty in detecting smaller boulders and overlapping craters. Insights gained from this analysis informed further model refinement to improve overall accuracy and robustness [6].

8) Validation: Model validation plays a critical role in assessing the performance of YOLOv11 and YOLO-NAS for lunar crater and boulder detection. Key aspects include:

- **Performance Metrics:** The following metrics were used to evaluate model effectiveness:
  - **Mean Average Precision (mAP):** mAP measures the overall accuracy of the model by averaging precision values at various Intersection over Union (IoU) thresholds, providing a comprehensive performance overview [7].
  - **Precision:** Indicates the proportion of correctly identified objects out of all detected objects, reflecting the model's ability to minimize false positives.
  - **Recall:** Represents the proportion of actual objects detected by the model, measuring its ability to minimize false negatives.
  - **Accuracy:** Represents the proportion of predicted objects that match the actual objects present, providing an overall assessment of model correctness.
- **Validation Dataset:** A separate subset (15% of the total dataset) was reserved for validation. This ensures unbiased performance evaluation during training.
- **Evaluation Process:** Model outputs were compared against ground truth labels to calculate the metrics. Bounding box predictions were assessed based on IoU thresholds (e.g., 0.5) to determine correctness.
- **Error Analysis:** Missed detections and false positives were analyzed to identify model weaknesses, such as difficulty detecting smaller boulders or overlapping craters. Insights from this analysis informed further model refinement [7].

9) Loss Optimization: Effective loss optimization is essential for ensuring the YOLOv11 and YOLO-NAS

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models accurately detect and classify craters and boulders. The training process incorporates a composite loss function to balance different types of errors:

- **Localization Loss:** - Measures the difference between the predicted and ground truth bounding box positions. - Encourages precise object localization by penalizing incorrect bounding box coordinates [13].
- **Confidence Loss:** - Evaluates the model's certainty in predicting the presence of an object within a bounding box. - Reduces false positives by penalizing incorrect high-confidence predictions for background areas.
- **Classification Loss:** - Ensures accurate object type predictions (e.g., crater, boulder) within the detected bounding boxes. - Penalizes incorrect class predictions and helps the model differentiate between object types.
- **Balancing Weights:** - The composite loss function assigns appropriate weights to localization, confidence, and classification components [13]. - This balancing prevents overemphasis on one type of error, ensuring overall model robustness.
- **Adaptive Optimization:** - During training, learning rates are dynamically adjusted based on the convergence of the loss components. - Early stopping mechanisms are used to avoid overfitting and ensure loss stabilization across validation datasets.

10) Pipeline Integration: The final model was integrated into the pipeline, where metadata alignment with spatial coordinates was applied to enable accurate object localization on lunar imagery [14].

IV. Results and Discussion

The evaluation results for YOLOv11 and YOLO-NAS are shown in Table I for the combined crater and boulder dataset, Table II for the crater dataset, and Table III for the boulder dataset. YOLO-NAS achieved 95

V. Model Performance Analysis

TABLE I
PERFORMANCE OF MODELS ON CRATER AND BOULDER DATASET

Model	Precision	Recall	MAP	Accuracy	Inference
Yolov11n	57.6	49.1	50.6	73	45ms
Yolov11L	59	50.2	51.1	74	113ms
YoloNASs	58.2	49	50.4	73	53ms
YoloNASl	60	51	52	75	123ms

A. Trade-offs Between Speed and Accuracy

As shown in Table II, YOLO-NAS achieved higher accuracy and precision in crater detection compared to YOLOv11, making it ideal for detailed analysis.

TABLE II
PERFORMANCE OF MODELS ON CRATER DATASET

Model	Precision	Recall	MAP	Accuracy	Inference
Yolov11n	75	72	79.8	88	30ms
Yolov11L	81	72.3	80.7	90	98ms
YoloNASs	84	74	81	91	35ms
YoloNASl	86	76	82	95	105ms

TABLE III
PERFORMANCE OF MODELS ON BOULDER DATASET

Model	Precision	Recall	MAP	Accuracy	Inference
Yolov11n	70.4	13.7	33.4	78	32ms
Yolov11L	72.3	14.7	35.2	79	100ms
YoloNASs	71.6	14	34	79.8	39ms
YoloNASl	73.4	16	40	82	108ms

Table III highlights the challenges in boulder detection, where smaller sizes and irregular shapes impacted recall. For the combined crater and boulder dataset, Table I demonstrates that YOLO-NAS outperforms YOLOv11 in overall accuracy but requires longer inference times. YOLOv11n, on the other hand, provides faster inference, making it suitable for real-time applications (see Fig. 4).

VI. Conclusion

This research automated the detection of lunar craters and boulders from high-resolution OHRC images captured by ISRO's Chandrayaan-2 Orbiter High-Resolution Camera using state-of-the-art object detection models, YOLOv11 and YOLO-NAS. The key findings are summarized below:

- **Effectiveness of YOLO Models:** Both YOLOv11 and YOLO-NAS demonstrated their ability to detect craters with high accuracy. The Neural Architecture Search (NAS) feature in YOLO-NAS significantly enhanced its adaptability to high-resolution datasets.
- **Detection Performance:** YOLO-NAS outperformed YOLOv11 in mean Average Precision

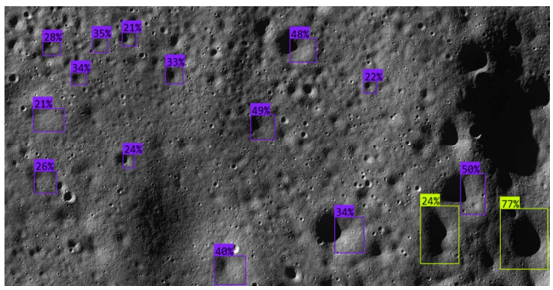


Fig. 4. Visualization of Detected Craters and Boulders

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(mAP) and recall rates, showing greater adaptability to complex and diverse lunar surface features.

- **Challenges in Boulder Detection:** Detecting boulders was more challenging due to their smaller sizes, irregular shapes, and overlaps with other features. Further pipeline refinements and hyperparameter tuning are needed to enhance detection rates.
- **Scalability and Computational Efficiency:** The pipeline efficiently processed OHRC images through techniques like image slicing, normalization, and GPU acceleration. These optimizations ensured scalability despite the computational demands of handling large datasets.
- **Scientific Implications:** The automated mapping of craters contributes to lunar geology by providing insights into impact history, surface evolution, and resource distribution. Integrating detected features with selenographic coordinates offers a foundation for future exploration missions.

Opportunities for Future Work:

- Developing real-time detection capabilities for dynamic mission needs.
- Extending the framework to other celestial bodies, such as Mars or asteroids, using planetary mission datasets.
- Refining the pipeline for better detection of smaller or overlapping features, such as boulders and subtle craters.
- Incorporating 3D modeling and elevation data for enhanced feature recognition and analysis.

Limitations:

- Detecting smaller boulders is challenging due to their irregular shapes and overlap with other features.
- YOLO-NAS has longer inference times, which may restrict real-time applications.
- Dependency on high-quality annotations for training, making the process resource-intensive.

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